
Integrable Systems — Exercise Sheet 13

The Yang–Baxter equation

Variant: Questions only

Conventions, as in Lecture 13: P is the permutation of the two factors of $\mathbb{C}^d \otimes \mathbb{C}^d$, $P(v \otimes w) = w \otimes v$; $\Pi_{\pm} = \frac{1}{2}(\mathbb{1} \pm P)$; $S(k) = (k - ic)/(k + ic)$, $c > 0$; $u = (k_1 - k_2)/c$; the rational R-matrix is $R(u) = u - iP$. On $(\mathbb{C}^d)^{\otimes 3}$, P_{ab} exchanges factors a and b , and X_{ab} acts on factors a, b (first leg on a). The Yang–Baxter equation is $X_{12}(u_{12})X_{13}(u_{13})X_{23}(u_{23}) = X_{23}(u_{23})X_{13}(u_{13})X_{12}(u_{12})$ with $u_{ab} = (k_a - k_b)/c$, so that $u_{13} = u_{12} + u_{23}$.

Exercise 1 (the permutation and its projectors). (i) Show that $P^2 = \mathbb{1}$, and that $\Pi_{\pm}$ are projectors with $\Pi_+ \Pi_- = 0$ and $\Pi_+ + \Pi_- = \mathbb{1}$. (ii) Show that $\text{tr } P = d$, and deduce the dimensions of the symmetric and antisymmetric subspaces. (iii) For $d = 2$, identify the two subspaces with the spin triplet and singlet.

Exercise 2 (Yang’s matrix). Two particles with labels in $\mathbb{C}^d$ and a label-blind delta interaction, with $\psi(x_1, x_2) = \pm P\psi(x_2, x_1)$ and, on $x_1 < x_2$, $\psi = Ae^{i(k_1 x_1 + k_2 x_2)} + A'e^{i(k_2 x_1 + k_1 x_2)}$. Lecture 13 shows that where ψ is symmetric in the positions $A' = S(k)A$ with $k = k_1 - k_2$, and where it is antisymmetric $A' = -A$. (i) For bosons the symmetric part is Π_+ : show that $Y_B = S\Pi_+ - \Pi_- = \frac{kP - ic}{k + ic}$. (ii) For fermions show that $Y_F = S\Pi_- - \Pi_+ = -\frac{kP + ic}{k + ic}$. (iii) For spin- $\frac{1}{2}$ fermions, what does Y_F do in the triplet and in the singlet? Why do two fermions with parallel spins not feel the delta?

Exercise 3 (along the momentum lines). (i) With $k = cu$, show that $X_B = PY_B = R(u)/(u + i)$ and $X_F = PY_F = R(-u)/(u + i) = -(u + iP)/(u + i)$. (ii) Show that $R(u)R(-u) = -(u^2 + 1)\mathbb{1}$, and deduce unitarity, $X(u)X(-u) = \mathbb{1}$, for both.

Exercise 4 (the Yang–Baxter equation for $u - iP$). On $(\mathbb{C}^d)^{\otimes 3}$: (i) Show that $P_{12}P_{13} = P_{13}P_{23} = P_{23}P_{12} =: C$, that $P_{13}P_{12} = P_{12}P_{23} = P_{23}P_{13} = C^{-1}$, and that $P_{12}P_{13}P_{23} = P_{13} = P_{23}P_{13}P_{12}$. (ii) Expand $R_{12}(u_{12})R_{13}(u_{13})R_{23}(u_{23})$ and $R_{23}(u_{23})R_{13}(u_{13})R_{12}(u_{12})$ by the number of P ’s, and show that they are equal exactly because $u_{13} = u_{12} + u_{23}$ (for $d \geq 2$, where $C \neq C^{-1}$). (iii) Deduce that X_B and X_F satisfy the Yang–Baxter equation.

Exercise 5 (a harmless-looking variant). Let $R^{\epsilon}(u) = u - i(1 + \epsilon u)P$. (i) Repeat the expansion of Exercise 4, and show that left minus right is $-\Delta(C - C^{-1})$ with $\Delta = \epsilon u_{12}u_{23}(2 + \epsilon u_{13})$. (ii) Deduce that R^{ϵ} fails the Yang–Baxter equation for every $\epsilon \neq 0$ (and $d \geq 2$); give Δ at $\epsilon = 0.2$, $u_{12} = 0.7$, $u_{23} = 1.3$.

Exercise 6 (the hexagon, for three labelled particles). Lecture 13’s three-body theorem puts the six amplitudes of the factorised ansatz on the permutations of S_3 and the contact conditions on the edges $\sigma \rightarrow \sigma s_j$. (i) Draw the hexagon, and list the two routes from the incoming wave (id) to the outgoing one (w_0), with the crossings each makes. (ii) (Computational.) For $d = 2$, $u_{12} = 0.7$ and $u_{23} = 1.3$, build X_B on $(\mathbb{C}^2)^{\otimes 3}$ as 8×8 matrices and compare the two products. (iii) Do the same for $R^{\epsilon}(u)/(u + i)$ at $\epsilon = 0.2$, and account for the size of the difference. (iv) Why is the equation automatic for particles without labels ($d = 1$)?

Exercise 7 ★. *The Yang–Baxter equation for $f\mathbb{1} + gP$.* Let $R(u) = f(u)\mathbb{1} + g(u)P$ with f, g continuous and g never zero, and $d \geq 2$. (i) Show that R satisfies the Yang–Baxter equation, with $u_{13} = u_{12} + u_{23}$, if and only if $h(u_{12}) + h(u_{23}) = h(u_{12} + u_{23})$ for all u_{12}, u_{23} , where $h = f/g$. (ii) Show that the continuous solutions of this equation are $h(u) = \kappa u$, and deduce that $R(u) = g(u)(\kappa u + P)$: for $\kappa \neq 0$, up to a scalar and a rescaling of u , the rational R-matrix. (iii) Check $u - iP$ against (ii).